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DBS' AI Journey

It was May 2024 and Piyush Gupta, Chief Executive Officer (CEO) of Singapore's largest bank, DBS, was engrossed in thought after watching OpenAI's online demo for the company's latest product, GPT-4o. Founded in 2015, OpenAI was a research company focused on the development of Artificial Intelligence (AI) technology for general purposes. OpenAI in 2022 released ChatGPT, a chatbot that produced high quality text and answers in response to user prompts and questions and proved that computers could mimic the complexities of human language and interactions.¹ At the core of ChatGPT was OpenAI's GPT-4, a type of large language model (LLM) used by computers to digest and create primarily text-based content.² LLMs fell under a subset of Generative Artificial Intelligence (GenAI), which referred to the use of AI systems to automatically generate new content of varying formats by learning from existing data.³ As an early adopter of GenAI technology,⁴ DBS had started experimenting with various applications based on earlier LLMs even before the release of GPT-4o and had seen promising, preliminary outcomes.

Compared to earlier versions of the technology, GPT-4o was able to handle additional input types like audio and video files.⁵ This marked a major step forward as machines powered by GPT-4o could factor into their responses contexts and subtleties better and produce more nuanced and natural outputs compared to machines that ran on GPT-4 and other older LLMs.⁶ While GPT-4o's potential could help DBS realize more revenue, its nascent stage required the bank to determine specific use cases and establish governance features to fully leverage its capabilities. Piyush was in a prior meeting that celebrated over DBS' record Q1 results announced less than two weeks before.⁷ It was widely recognized within DBS that the profit increase was partly attributed to its AI initiatives.⁸

DBS' foray into AI started in 2014 with an initial experiment that ended abruptly when the results fell short of expectations. The bank continued with the AI experimentations, however, and partnered with other organizations to refine its approach. At the same time, DBS was undergoing a digital transformation, known internally as the "Digitalization Wave," to become as agile and innovative as a technology start-up.⁹ Part of the bank's efforts to digitalize was to build up capabilities in data analytics, which focused on reviewing historical data for patterns and making predictions.¹⁰ In 2018, DBS established the P-U-R-E framework—an acronym for Purposeful, Unsurprising, Respectful, and Explainable—as an internal guideline to ensure proper governance and ethical standards in data use. P-U-R-E was subsequently expanded to encompass the use cases of AI within the bank.¹¹ The

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digitalization wave also prompted a reorganization of teams and workflows around technology and customer needs, enhancements in data architecture and governance, and the reskilling and upskilling of existing talent. The digital transformation helped lay the foundation for DBS' later work around AI, which took data analytics to a new level.¹²

The advent of GenAI in late 2022 prompted the bank to incorporate the latest technology in DBS' business activities. Bank employees saw the creation of powerful digital tools used internally that increased overall productivity and elevated customer experiences. Despite its promises, GenAI introduced new complexities, necessitating DBS to adapt its existing operational processes and governance framework to effectively manage and leverage this emerging technology. Specifically, the bank had to supplement the existing P-U-R-E framework with increased human oversight to monitor the potential risks brought about by GenAI. Piyush and his team identified four specific business areas that DBS later focused its GenAI efforts on, instead of experimenting broadly as they did with AI.

With the monthly meeting of DBS' GenAI Steering Committee in sight, Piyush wondered how he and his team could better steer the bank toward harnessing the potential of GenAI technologies more safely and responsibly.

Background

Established in July 1968, DBS, originally known as the Development Bank of Singapore, had a core mandate to finance the country's industrial and urban development. Over the years, the bank successfully led projects that helped grow its position as one of Southeast Asia's^a biggest banks by assets. Some notable ones included the financing of prominent real estate developments like Plaza Singapura (1975) and Raffles City (1986), as well as the IPOs^b of Singapore Airlines (1985) and Singapore Telecom (1993).¹³ The bank expanded its domestic and overseas businesses and acquired Singapore's Post Office Savings Bank (1998), Hong Kong's Dao Heng Bank (2001), Taiwan's Bowa Bank (2008), and several others in the region, and was the first Singaporean bank to incorporate in China (2007).¹⁴ It rebranded as DBS Bank Limited in 2003 to reflect its regional presence.

In 2023, DBS reported total assets of US\$560 billion, total income of US\$15 billion, and net profit of US\$7.5 billion.¹⁵ (See **Exhibit 1a** for summary of DBS' selected financials from 2021 to 2023.) DBS was publicly listed on the Singapore Exchange in 1999,¹⁶ and its largest^c shareholder was Singapore's state-owned investment arm, Temasek Holdings.¹⁷ The bank was given "AA-" and "Aa1" credit ratings by Standard & Poor's and Moody's, which were among the highest in the region. DBS' three core operations were in Institutional Banking, which was the largest contributor to DBS' total income,¹⁸ Consumer Banking and Wealth Management, and Treasury Markets.¹⁹ (See **Exhibit 1b** on DBS' key financials by business segment from 2021 to 2023.)

In the early 2000's, DBS suffered in terms of customer satisfaction scores.^{20, 21} It was known then to be "Damn Bloody Slow," a dig at the bank's reputation for being "bureaucratic, unimaginative and

^a DBS' presence in Southeast Asia: Singapore, Indonesia, Malaysia, Myanmar, Philippines, Thailand, Vietnam. DBS, "Our network," <https://www.dbs.com/in/corporate/why-us/our-network>, accessed March 2024.

^b Initial Public Offerings.

^c As of February 7, 2024, Temasek held over 29% of DBS' shares. DBS, "Annual Report 2023," <https://www.dbs.com/iwov-resources/images/investors/annual-report/dbs-annual-report-2023.pdf?pid=sg-group-pweb-investors-annual-report-2023-pdf-2023>, accessed March 2024.

unresponsive.”²² Such were the circumstances when Piyush, then 49, took over as the CEO in 2009.²³ In the years that followed, Piyush led DBS through a series of radical transformations, informed by his previous experience in technology and operations within banking, and a short stint as a technology entrepreneur.²⁴ The bank went on to win prestigious titles such as “Global Bank of the Year” and “World’s Best Digital Bank” for 2021, and “Best Bank in the World” for 2022.²⁵

AI as a New Technology

The term “artificial intelligence” first emerged in the 1950s and represented a type of technology aimed at simulating human intelligence and problem-solving capabilities.²⁶ AI’s ability to ingest large amounts of data, analyze it for correlations and patterns, and make predictions about future states presented a wide array of applications in different areas, from daily life to commercial operations.²⁷ Advances in other essential technologies like data storage, networks and semiconductor chips especially in the 2010s helped lower the costs of AI.²⁸ While AI brought numerous benefits like enhanced digital capabilities and increased customer engagement in the context of banking,²⁹ not all banks showed the same level of urgency in adopting the technology.³⁰ One reason was that banks and financial institutions were typically subjected to strict regulatory oversight.³¹ Another reason had to do with outdated, legacy banking systems that were costly to upgrade and replace.³² Although DBS’ initial efforts to stay at the forefront of technological developments were not specifically aimed at AI, these efforts equipped the bank with the capabilities to scale AI when the right opportunity came about.

AI at DBS

DBS’ first formal encounter with AI started in 2014 when the bank signed up with IBM^d to experiment on its Watson computer system.³³ Watson was a cloud-based computing system capable of processing vast amounts of information and learning from each interaction at unprecedented speeds.³⁴ Piyush’s vision then was to use Watson to process finance-related information from Bloomberg and generate advice on wealth management for the bank’s clients on a real-time basis.³⁵ The initial experiment did not yield the desired outcome and Piyush realized he was “overly optimistic” then about Watson’s natural language processing (NLP) capabilities—a subset of AI used in the processing of human language by a computer program to perform tasks like text translation, sentiment analysis and speech recognition.³⁶ The NLP that Watson used in its earlier days was limited to breaking down sentences and groups of words, and was not capable of complex tasks like executing commands given by its user.³⁷ As Piyush recalled, Watson at that stage “did not understand contexts well enough, was unable to read PDFs^e and graphs, and had weak NLP and predictive capabilities.”³⁸ The experiment ended after 18 months when Watson’s predictions achieved only around 80% accuracy, which Piyush concluded was “not good enough to take to market.”³⁹

In 2014, the bank set up a joint laboratory with Singapore’s Agency for Science, Technology and Research (A*STAR) to focus on data analytics and other innovations for the banking industry.⁴⁰ In Piyush’s own words, the joint effort allowed DBS to “dirty its hands” and created several AI models that yielded some outcomes. In 2016, the bank invested in Kasisto, a consumer banking startup venture that helped develop DBS’ NLP capabilities for Asian languages.⁴¹ The venture produced a digital chatbot that tended to make small but significant mistakes when dealing with open-ended questions and answers, according to Piyush’s observations.⁴² In a tightly regulated sector like banking, even

^d International Business Machines Corporation.

^e Portable Document Format.

having a marginal error rate was not good enough.⁴³ As a result, DBS transitioned to what Piyush termed “guided conversations.” In this approach, the bank’s customer service officers asked the chatbot closed-ended questions and then used the responses generated to address customer queries more effectively.⁴⁴ This reduced the potential for errors and proved a more effective way to harness AI in creating value for the bank and its customers.

In the subsequent few years, Piyush and his team continued to explore in the field of AI. By 2020, they began to focus on applying machine learning (ML) – a branch of AI that used data and algorithms to enable machines to learn from experience and improve, much like humans – to data analytics.⁴⁵ The bank also centered its AI efforts around the use of structured data, which typically consisted of clearly defined, tabular information organized into rows and columns.⁴⁶

Organizational Structure

Transformation Group

Formed in 2009, the Transformation Group (TG) was to help drive the bank’s transformative programs. As part of DBS’ Technology and Operations department, the TG’s core function was to identify and monitor external developments that had implications for DBS, assess whether change was necessary in response to these developments, and put together a plan to build the various essential components for the change.⁴⁷ Nimish Panchmatia, Chief Data and Transformation Officer, explained that instead of performing the transformation on its own, the TG played a catalytic change function tasked with empowering DBS employees “to implement transformation as part of their workflows and processes so that it is more embedded within the day to day.”⁴⁸

A pivotal moment came in 2012 when Piyush met with Jack Ma, founder of Chinese tech giant Alibaba and leading digital payments platform Alipay, at the company’s headquarters in Hangzhou, China. The encounter spurred Piyush to step up DBS’ digital transformation,

I realized at that stage that the fintech companies were completely reimagining the future of our industry. Because they were learning how to acquire customers without physical presence and how to do transactions without paper. They were learning about and beginning to do instant fulfillment. They were already using data to do credit underwriting. And it became quite clear to me and my team at that time, that this was going to be the future. So, we took a big leap of faith at that time. We were empowered by a board. It was very critical.⁴⁹

In the quest to benchmark DBS against technologically advanced companies, Piyush and his team coined the acronym GANDALF^f—representing Google, Apple, Netflix, DBS, Amazon, LinkedIn, and Facebook—positioning the bank alongside these exemplars.⁵⁰ The TG implemented a bank-wide digital transformation program organized around three pillars: a) the redesign of DBS’ technology architecture and shift to a cloud-based IT infrastructure; b) renewed focus on “customer-obsession” as a way to measure and drive outcomes; and c) the embracing of start-up culture to remain nimble, data-driven and unafraid of taking risks.⁵¹, ⁵² To Piyush and his team, carrying out bank-wide transformations was not only necessary to stay relevant and to keep up with evolving customer needs;

^f Gandalf was the name of a wise wizard from J.R.R. Tolkien’s *The Lord of the Rings* and *The Hobbit* novels.

a well-designed transformation program was a good way to galvanize employees and ensure there were no “first-class and second-class citizens” in the process.⁵³

From Business and Technology to Managing Through Journeys

The rise of technology-focused companies in the early 2010s showcased to the world a new way of leveraging technology to make informed business decisions.⁵⁴ The key takeaway for DBS was for the business and technology teams to interact regularly and for every decision to be weighed equally from both perspectives, as opposed to the previously disconnected way of working on projects separately. This led to the 2018 establishment of platforms, based on the “two-in-a-box” concept, where business and technology units assumed joint ownership of projects.⁵⁵

The bank then implemented the “horizontal organization (HO),” bringing cross-functional teams together virtually to work on common projects to enhance customer experiences and drive value for the bank.⁵⁶ Under the HO, cross-functional teams collaborated on projects and became more embedded in each other’s work, allowing the bank to adapt nimbly to challenges—such as when DBS had to quickly step up its digital offerings during the COVID-19^g pandemic.⁵⁷

In 2021, building on the insights from HO, Piyush and his team created Managing through Journeys (MtJ). MtJ focused on delivering business outcomes in the customer’s end-to-end journey by applying the HO approach more broadly across various teams involved in the customer journey value chain (including sales and marketing, product design, fulfillment in operations, compliance, and risk management).^{58, 59} For instance, the bank’s Treasury Markets, Global Transaction Services, and technology teams collaborated horizontally to transform DBS’ existing foreign exchange (FX) business, leading to increased responsiveness to customer needs and reduced turnaround times for new product and service innovations. Without MtJ, improving a customer journey such as FX could have required a more tedious process of coordinating across multiple verticals: Treasury Markets might define customer pain points, Global Transaction Services might assess the implications for payments and liquidity, technology teams might build or modify systems, operations teams might handle fulfillment, and compliance and risk teams might review controls. The process could therefore have depended on a series of handoffs, approvals, and reconciliations across units with different priorities and metrics. MtJ was designed to reduce this friction by organizing these parties around the same journey, dashboard, KPIs, and accountability structure. These improvements led to a 15% revenue growth and an overall customer satisfaction score of 4.47 (out of five) for the FX business in 2022.⁶⁰ Another example was the Retail, Commerce, and Services (RCS) MtJ that brought together teams across various functions like corporate banking, consumer banking, and operations and technology, to deliver digital payment and collection services that minimized friction for the bank’s customers.⁶¹

MtJ were run using a value map that established relationships between defined levers, drivers and outcomes for customer journeys. Each MtJ operated as a single performance cell,^h comprising various “squads” or sub-teams headed by “mini-CEOs” who oversaw specific projects. (See **Exhibit 2** on the make-up of a typical performance cell.) Under MtJ, the different teams also used the same dashboard to track their shared budgets, KPIsⁱ and accountability, and had the common objective of delivering customer, financial, employee, and risk and control outcomes.⁶² (See **Exhibit 3** on the different

^g Coronavirus Disease. During its peak, the pandemic lasted for about three years between 2020 and 2022.

^h Performance Cells were cross-functional teams organized horizontally and aligned on common KPIs.

ⁱ Key Performance Indicators.

elements within MtJ.) By May 2024, DBS had implemented close to 60 MtJs across the bank, involving approximately 400 leaders across these MtJs serving as mini-CEOs.

Data Architecture

Advancing DBS with AI (ADA)

In 2018, DBS operated with 12 data warehouses supporting its operations. While data was shared across various teams, each team ran its own data operations, with limited opportunities to reuse analytical models developed by others.⁶³ The fragmented approach to data use and siloed processes resulted in duplicated efforts. This prompted the consolidation of DBS' data and its migration to an end-to-end integrated central data platform which stored all the data, analytics, and data science processes for building and deploying analytical solutions, and enabled virtual pooling—allowing data to be stored in legally required locations while maintaining seamless access across jurisdictions.⁶⁴ The idea was for the central data platform to serve as the single “source of truth” for the bank's data operations.⁶⁵

Additionally, DBS in 2018 established an Analytics Center of Excellence to scale advanced analytical capabilities throughout the bank. This central data team operated on a federated model, embedding some members within various business units to respond more effectively to specific needs.⁶⁶ The central team focused on more advanced tasks such as neural network modeling, while analysts in the individual business units handled essential but more straightforward tasks such as running regressions.⁶⁷ This structure enabled the central team and its sub-teams to specialize in areas like data management, which involved the collection, organization, and storage of DBS' data for business analysis, and in data analytics, which focused on cleaning and interpreting this data to support decision-making within the bank. Together, these specializations helped accelerate work processes.

The central data platform was later named Advancing DBS with AI, or ADA, after Ada Lovelace, an English mathematician and programmer from the 18th century. ADA featured a data “backbone” that consisted of DBS' core data sets from all its units, a “laboratory” that functioned like a sandbox for internal experimentations, and a “factory” that promoted data products and services that were operationally ready.⁶⁸ The three components were critical in scaling the bank's data operations.

Data in ADA was organized into three distinct tiers: tier one (T1) contained clean and validated data from various system sources; tier two (T2) was made up of reusable data sets built around products, transactions and other operations of the bank; and tier three (T3) housed data sets for specific use cases that were non-reusable and restricted to only relevant teams.⁶⁹ All data on ADA was made discoverable and traceable with proper metadata^j and lineage^k.

Most importantly, data access was automatically granted pre-approval based on the user's job and role, which meant that employees did not have to go through lengthy approval processes before using the data.⁷⁰ (See **Exhibit 4** on how data access was granted based on job.) For example, a data scientist in the consumer banking department could instantly access all consumer banking-related datasets on the platform, a process that previously took months.⁷¹

^j Metadata was data that provided information about other data. It described, explained, located, and made it easier to retrieve, use, or manage data resources.

^k Data lineage tracked how data flowed from source to consumption—origins, movements, transformations, and destination.

Additionally, DBS had a bank-wide technology forum that tracked the relevant KPIs across all units to promote the use of ADA and monitor its rate of adoption. One of the KPIs measured was the percentage of their existing use case-specific data sets that the business units transformed into reusable ones.⁷² The bank also took care to not lock itself in with external technology vendors. DBS adopted a hybrid architecture that consisted of on-premise data centers and cloud-based services to host all of its data. The architecture also prioritized a modular design to allow for flexibility in switching out individual components of the data platform.⁷³ DBS also did not shy away from building in-house solutions and invested heavily on its own technological capabilities in the absence of suitable external service providers.⁷⁴ The result was a platform incorporating the “best of breed” solutions, tailored to meet DBS’ specific requirements.⁷⁵

ALAN

In 2021, DBS created ALAN, a system that was a key component of DBS’ efforts to deploy AI throughout the bank.⁷⁶ The name “ALAN” was inspired by Alan Turing, the father of theoretical computer science and AI. ALAN contained the key documentation on all the bank’s data, use cases, and models. It served as a “library” that employees referenced when working on new projects, and as a repository that kept track of published models and best practices.⁷⁷ Petri Tuomola, Head of Data Platforms, illustrated: “You can go to ALAN and say, ‘I want to find anyone who’s done a credit card attrition problem or figured out a way to sell complementary products to customers,’ and you could find the source code, the data sets they used, the model performance metrics, the model deployments, etc., all from ALAN.”⁷⁸ (See **Exhibit 5** for summary on key capabilities of ADA and ALAN.)

As a result of having ADA and ALAN, the timeline for AI model development and deployment decreased from between 12 to 15 months in 2018 to between two to three months in 2023. (See **Exhibit 6** for comparison of specific timelines pre- and post-ADA and ALAN.) This also increased AI adoption rates across the bank. Between 2018 and 2024, around 350 use cases were created bank-wide, informed by approximately 800 models, with the numbers continuing to grow each month.⁷⁹

The creation of ADA and ALAN was an important milestone in DBS’ AI industrialization. As Piyush recalled, one of AI’s earlier applications was in predicting employee attrition and facilitating internal mobility.⁸⁰ Another example was with the DBS NAV Planner, a mobile application-based digital tool that provided “hyper-personalized” financial planning services to the bank’s retail customers for free. (See **Exhibit 7** for screenshots of NAV Planner.) As Tse Koon Shee, Head of Consumer Banking and Wealth Management, explained, the mechanics involved monitoring a customer’s financial health and sending out appropriate digital nudges to help them foster healthier spending habits.⁸¹ For example, the bank found that compared to non-users, customers who actively used the NAV Planner saved 83% more, were 4.3 times more likely to invest their increased savings, and were 2.3 times more likely to have purchased insurance.⁸²

In 2022, Piyush and his team began quantifying the additional value generated by integrating AI into the bank’s activities. For each use case, a control group that did not use AI was established, and its outcomes were compared against those of the experimental group that used AI. The differences captured were then attributed to AI.⁸³ The method identified approximately SGD80 (US\$59) million,⁸⁴ SGD150 (US\$111) million, and SGD370 (US\$274) million in economic value attributed to the use of AI

alone in 2021, 2022, and 2023 respectively, and allowed for a projected SGD700^l (US\$519) million for 2024.⁸⁵

As Panchmatia pointed out, DBS was not afraid of failures either, and used every opportunity to improve existing AI efforts, “We’re very careful about ensuring that we see the benefits. We’re quite good at killing models that we believe do not provide some form of an ‘uplift’ to the business or benefit to our customers and employees. In those instances, we’ll ask ‘what went wrong?’ We made our assumptions, we went with it, but it didn’t work. Sometimes it means there was something we missed, and that in itself is good learning as well.”⁸⁶

Governance

DBS participated in broader discussions around responsible AI use. In 2018, the bank contributed to the Monetary Authority of Singapore’s FEAT^m principles, designed to guide the responsible use of data in the country’s banking sector. In the same year, DBS introduced its own P-U-R-E framework that was applied to all data use cases within the bank.

In 2020, the bank developed an internal safeguard known as the Responsible Data Use (RDU) model, aiming to streamline processes, minimize bureaucracy, and empower employees.⁸⁷ The RDU model consisted of three components: 1) alignment with broader regulatory frameworks around data management; 2) the bank’s own P-U-R-E framework on data-use ethics; 3) model governance that focused on the ongoing development and implementation of additional guardrails for emerging technologies such as AI.^{88, 89}

P-U-R-E

DBS’ P-U-R-E framework sought to address the subjective nature of data-use ethics.⁹⁰ As Sameer Gupta, Chief Analytics Officer, summarized,

We need to be clear about the **Purpose** of our use of data. Then we need to be **Unsurprising** to our customers and stakeholders, even if there is a legitimate purpose. The data also needs to be used in a **Respectful** and responsible manner—and that includes how we collect it and communicate to our customers on how we use the data. The last criterion is about how **Explainable** the model and data outputs are internally and to the customers.⁹¹

In Panchmatia’s words, P-U-R-E was aimed at giving “the right level of empowerment to people who were doing data and AI-related work on a day-to-day basis,” and at the same time creating guardrails to ensure that the bank’s principles were not compromised.⁹² P-U-R-E was governed by the RDU Committee, which consisted of around 20 senior leaders from cross-functional teams that examined the data employed, use cases,ⁿ and the way communication was done around data-use by DBS. In a process that was formalized into a procedure known as “white-listing,” AI use cases meeting all aspects of P-U-R-E were approved for deployment and documented. For use cases similar to those

^l The additional economic value for 2024 was projected by DBS at the time of writing.

^m Fairness, Ethics, Accountability and Transparency.

ⁿ In this context, a use case referred to a specifically defined application of data or AI technology to solve a business problem or to achieve a certain objective.

already white-listed, teams could complete a questionnaire to confirm compliance with P-U-R-E, avoiding the need for a full RDU Committee review.⁹³ An example of this white-listing process was the bank's Next Best Conversation for Wealth Customers initiative, which used data insights to provide relationship managers with intelligent, personalized prompts for customer engagement.⁹⁴ The owner of this use case completed a self-assessment against the P-U-R-E principles to ensure customer data was handled and used responsibly. Upon confirmation that the use case adhered to these principles, it was submitted to the designated approver for final review and approval. Once approved, the use case was added to the white-list, allowing similar future applications to follow a streamlined approval process. Su Shan Tan, Deputy CEO and Head of Institutional Banking, elaborated on the usefulness of P-U-R-E in guiding employees on AI use: "You have to remember what P-U-R-E stands for. Once you have that in your mind, then it becomes the minimum standard. Every time you do something AI-related, you ask, 'is this P-U-R-E?' Then at least you know the bases are covered."⁹⁵ This approach enabled employees to innovate with AI and scale its implementation much more quickly.⁹⁶

Talent

Career Development

In response to a global shortage of technology talent, particularly in data science and artificial intelligence, DBS recognized the need to develop its own talent pool to drive its data and technology initiatives. In 2023, the bank had on its payroll around 2,500 data scientists, analysts and engineers, and had roughly twice as many technologists as bankers.⁹⁷ This also highlighted the bank's efforts to benchmark against leading technology companies, offering data professionals technical career paths along which they could grow.⁹⁸ More broadly, DBS' total employee base grew from over 29,000 in 2020 to approximately 41,000 in 2024.⁹⁹

DBS also partnered with external organizations to expand its AI talent pool. Notably, in 2018 the bank launched the Analytics Capabilities Enhancement program, which was later renamed the Data Management Associates program, to recruit talent directly from Singaporean universities. The National University of Singapore was the inaugural partner for this initiative, which featured a tailored curriculum aimed at cultivating enterprise-level data science skills.¹⁰⁰ The program also sought out fresh graduates with backgrounds in data science and provided further on-the-job training and exposure to participants.¹⁰¹ The bank launched the DBS FinTech Apprenticeship in 2022 to equip polytechnic students with practical knowledge and industry experience, and offered the opportunity to join DBS as a permanent member of its technology operations team.¹⁰²

Continued Education and Training

DBS focused on upskilling and reskilling employees with the right knowledge in the age of AI. Piyush saw this as the most important piece and referred to it as educating the employees on the "art of the possible."¹⁰³ Amid the COVID-19 pandemic in 2020, DBS partnered with Amazon Web Services (AWS) to launch the DBS x AWS DeepRacer League, in which participants went through hands-on virtual training and programmed their own autonomous model race car.¹⁰⁴ This fostered excitement throughout the bank for learning about AI, fueling creativity, much like illuminating "light bulbs in the minds of the front-end users."¹⁰⁵ In 2022, DBS launched iGrow, an AI/ML-enabled digital career coach that helped employees identify future career aspirations, relevant skills, and training opportunities for upskilling or reskilling.¹⁰⁶

DBS was early to recognize the importance of data literacy—knowing the right questions to ask when solving data problems—in order to become truly “AI-fueled.”¹⁰⁷ DBS’ Data Chapter, a bank-wide structure created in 2023 to centrally develop, manage, and deploy data professionals across the organization, developed learning pathways to equip bank employees with relevant skills and tools to use AI-related techniques in work and become “AI-ready.”^{108, 109} For example, DBS’ employees participated in a recurring “Data Heroes” program to learn about the more technical concepts in AI and around data use. By the end of 2023, more than 9,000 employees benefitted from the learning pathways developed by the Data Chapter.¹¹⁰ As Sameer explained, continued learning was important to the bank and its people: “We are all in that same journey—getting educated on topics like data governance, responsible data use, etc. Our investment in education and fostering a culture of continuous learning will be a significant differentiator for us.”¹¹¹

The Introduction of GenAI

The release of ChatGPT by OpenAI in November 2022 ushered in a new phase in AI for the world and for DBS.¹¹² ChatGPT, an example of a Generative Pre-training Transformer (GPT), presented a new way for businesses to process unstructured data.¹¹³ The ability of the tool to generate human-like language responses with user input was what merited the broader term of GenAI. The new technology took things to another level with the ability to autonomously synthesize various types of content like images and texts from ever-larger amounts of data.¹¹⁴ This was made possible with breakthroughs in transformers, a type of ML algorithm that enabled the use of unlabeled data and enhanced connection-tracking across data groups. Advances in LLMs too allowed GenAI models access to huge numbers of data parameters to create hyper-realistic content.¹¹⁵ However, GenAI also introduced several new challenges, including a tendency to produce “hallucinations” resulting in false information and biases that reinforced undesirable stereotypes.¹¹⁶ To Piyush and the bank, the arrival of GenAI prompted a new way to realize what they set out to accomplish with NLP back in 2013,

With ChatGPT and GenAI, like everybody else, we thought that it could be quite a game changer. One is it takes NLP to a level where I think you could actually use unstructured data. All the progress we’ve made is mostly on structured data. So, my vision, which I had 10 years ago of plugging Watson into Bloomberg, is now realizable with the progress in NLP with GenAI. The second big thing is obviously the generative component. So far, all the AI that we use is for predictive purposes.¹¹⁷

Group Strategy, Transformation, Analytics and Research (GSTAR)

In July 2023, DBS unified the previously separate Group Strategy; Transformation; Analytics and Research teams to establish GSTAR.¹¹⁸ Him Chuan Lim, Head of GSTAR, explained that the new structure was designed to coordinate and align strategy execution across the bank’s various businesses while driving a comprehensive agenda. This agenda encompassed both the pursuit of GenAI and the broader industrialization of AI throughout DBS’ operations. In Lim’s words, GSTAR’s role was “to architect, catalyze, and drive the strategic transformation of DBS to be a differentiated bank, powered by research and AI, in a pragmatic way.”¹¹⁹

Enhanced Governance

DBS saw the need to adapt its governance framework as the risks became more complex with the arrival of GenAI. A primary area of concern was GenAI’s tendency to hallucinate and create nonsensical or even harmful outputs.¹²⁰ Hallucinations were in violation of the “Explainable” criterion

under P-U-R-E, and the complexity of the algorithms made it challenging for the team to fully understand the backend processes and trace the source of the outputs. As a result, DBS relied on always having a “human-in-the-loop” to ensure oversight during GenAI model development and deployment to mitigate the new risks.¹²¹ This meant that while the bank’s AI applications interacted with customers, the GenAI applications were restricted to internal employee use only and were not customer-facing. All GenAI use cases were also brought to the RDU Committee for in-depth review to identify inherent risks and to ensure adequate safeguards. In early 2023, the bank established a GenAI Steering Committee that looked at DBS’ efforts in GenAI specifically.¹²² For example, it was tasked to monitor emerging external risks and decide on corresponding mitigation strategies.¹²³ Lam also pointed to the importance of identifying the “incremental risks” associated with GenAI use, and in building up the technical capabilities to manage these risks: “otherwise, you end up with this technical complexity being administered by a small number of “experts”, which can be an inhibitor to scaled GenAI use.”¹²⁴

Areas of Implementation

Piyush and his team remained focused on delivering value for the bank and its customers while tapping into GenAI’s capabilities. Referring to the enthusiasm surrounding the potential of GenAI, Sameer highlighted DBS’ commitment to maintaining a realistic approach,

We’re being very cognizant that there are lots of unknown unknowns right now in the GenAI space but we’re looking to solve them. Because if your expectation is that it will deliver the correct results 100% of the time, then we’ve already lost the game. GenAI offers an unprecedented opportunity to unlock the power of unstructured data while providing our employees with powerful co-pilots to reduce toil. We should not lose sight of such game-changing potential.¹²⁵

In the first few months of 2023, around 20 of DBS’ key technology and business leaders convened during a one-day workshop to strategize on the future with GenAI. Instead of letting innovation and experimentation take place freely (which Piyush referred to as “letting a thousand flowers bloom”) as with other new tools, the approach toward GenAI experimentation would be more controlled. Drawing insights from relevant literature, the senior management devised a strategy they termed “one horizontal and three verticals,” which involved exploring opportunities across different areas of the bank’s activities.¹²⁶

The focus on the horizontal capability led to the development of the DBS-GPT, the bank’s proprietary GenAI-enabled digital assistant. DBS-GPT integrated multiple third-party foundation models, like GPT-4, BART, and Claude, into its backend, to provide features such as reading and synthesizing capabilities.^{127, 128} DBS also started the development of an Enterprise Knowledge Base (EKB) that was to be integrated into DBS-GPT to enable easy search and retrieval of unstructured and domain-specific knowledge within the bank’s context.¹²⁹ The goal was to increase the accuracy and relevance of DBS-GPT’s output and speed up internal tasks such as text-processing, insight-extraction and response generation that were traditionally labor-intensive. DBS-GPT exemplified what Piyush and his team called “inch-deep, mile-wide,” which referred to helping employees build a basic capability (“inch-deep”) in GenAI to cater to the diverse needs of a broad user-base (“mile-wide”).¹³⁰

The three vertical areas of opportunity were about “re-imagining workflow and the nature of work itself” and focused on using GenAI to augment: 1) call center operations, 2) relationship management, and 3) internal system development processes like coding and user-testing.¹³¹ DBS anticipated that by implementing GenAI as a ‘co-pilot’ to assist human operators in efficiently managing customer queries,

call centers could experience a 20% boost in productivity.¹³² The bank's relationship managers also used GenAI in areas that were notorious for causing employee "toil," such as preparing background research for client visits, or performing news screening for Anti-Money Laundering (AML) investigations and Know-Your-Customer (KYC) processes.¹³³

DBS in the Age of GenAI

Between 2009 and 2024, DBS reinvented itself and transformed from the "default salary-crediting bank" for civil servants and military personnel to a "Model Bank for AI Industrialization."^{134, 135} The bank was after something "bigger" than just recognition; as Panchmatia remarked, the vision was "to be the best for our customers, to be a great place to work at for our employees, and ultimately be the best bank for a better world. Having that centrality of purpose is what really drives us forward."¹³⁶ DBS experienced several disruptions in its digital services in 2023 but took the opportunity to identify gaps and deficiencies in change management, system resilience, incident management, and technology risk and governance.¹³⁷ The result was the implementation of a roadmap aimed at increasing DBS' technology resilience and oversight.

Lim drew from Sun Tzu's Art of War, emphasizing the significance of leadership and cultivating the right mindset throughout the organization in order to survive changes and thrive afterward. Having a clear purpose helped DBS overcome various obstacles in developing the right capabilities to compete and deliver value in an increasingly digital world. Shee highlighted that instead of putting on "digital lipstick," or chasing after superficial changes, DBS "integrated digitalization into its DNA" and differentiated itself from competitors.¹³⁸

Similarly, DBS could not afford to overlook the potential of the latest advances with GPT-4o. Piyush explained, "We don't have a choice. As the former Chairman of Citigroup, Chuck Prince, once said, 'when the music is on, you have to get on the dance floor.'"¹³⁹ He recognized that despite the potential benefits of allowing GenAI to interact more directly with customers, the bank could not ignore the associated errors that might ensue, no matter how rarely they might occur.¹⁴⁰ Piyush pondered whether the existing AI governance framework could adequately address the unique challenges posed by GenAI implementation and allow its usage to really scale with minimal human intervention. He also wondered how the pursuit of GenAI might affect DBS' workforce.

Exhibit 1a Summary of DBS' Selected Financials from 2021 to 2023

In US\$ millions	2021	2022	2023
Selected income statement items (excluding one-time items)			
Total income	10,499	12,211	14,933
Profit before allowances and amortization	5,712	6,965	8,972
Allowances for credit and other losses	38	175	437
Profit before tax	5,754	6,943	8,687
Net profit excluding one-time items	5,033	6,063	7,612
Selected balance sheet items			
Total assets	507,694	550,092	547,083
Customer loans	302,655	306,744	307,961
Total liabilities	464,986	507,859	501,020
Customer deposits	371,450	389,980	395,976
Total shareholders' funds	42,569	42,096	45,928
Selected financial ratios (%)			
Net interest margin	1.45	1.75	2.15
Cost-to-income ratio	45.60	43.00	39.90
Return on assets	1.02	1.12	1.38
Return on equity	12.50	15.00	18.00
Loan / deposit ratio	81.50	78.70	77.80

Source: DBS Annual Report 2023.

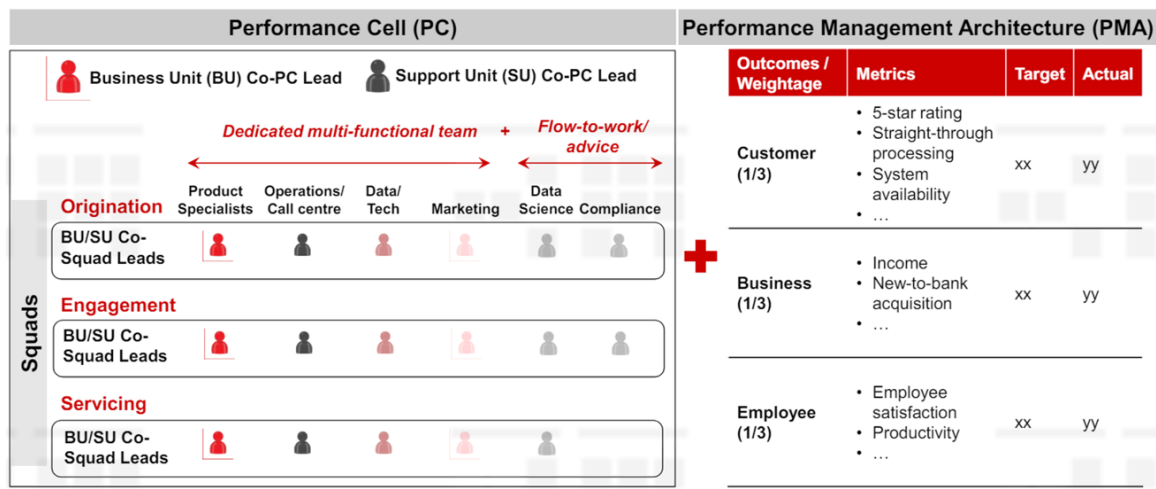
Note: Original figures were reported in Singapore Dollars (SGD) and have been converted into U.S. Dollars (US\$) based on the exchange rate: SGD1 = US\$0.74.

Exhibit 1b DBS' Key Financials by Business Segments from 2021 to 2023

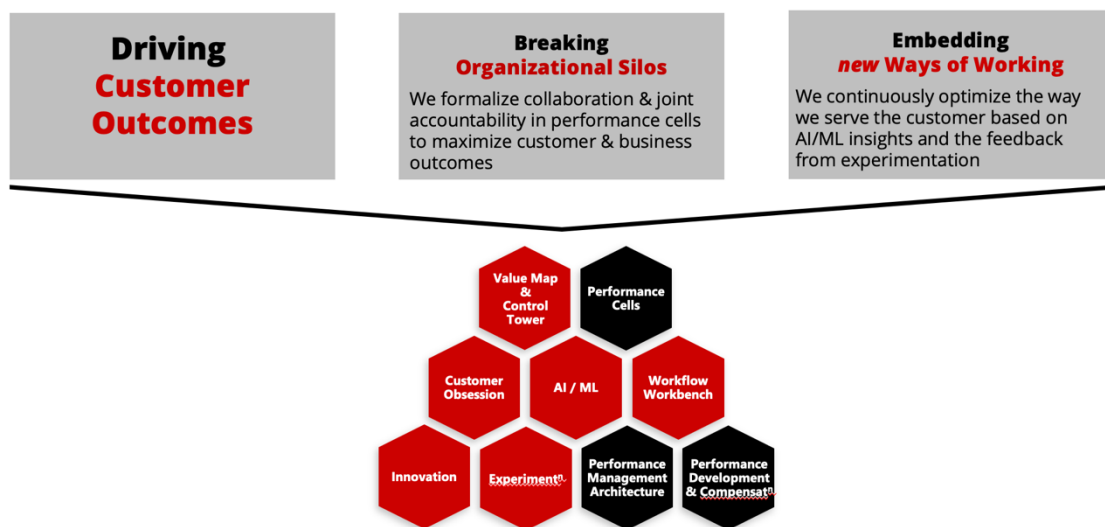
In US\$ millions	2021	2022	2023
Total income			
Consumer Banking/Wealth Management	3,938	4,924	6,615
Institutional Banking	4,428	5,689	6,924
Treasury Markets	1,117	869	537
Others	1,016	730	844
Total	10,499	12,211	14,920
Total expenses			
Consumer Banking/Wealth Management	2,481	2,814	3,265
Institutional Banking	1,544	1,668	1,842
Treasury Markets	479	458	466
Others	357	306	562
Total	4,861	5,247	6,135
Amortization of intangible assets			
Consumer Banking/Wealth Management	0	0	0
Institutional Banking	0	0	0
Treasury Markets	0	0	0
Others	0	0	7
Total	0	0	7
Allowances for credit and other losses			
Consumer Banking/Wealth Management	34	117	200
Institutional Banking	104	-151	65
Treasury Markets	-4	-7	11
Others	-96	217	161
Total	38	175	437
Share of profits or losses of associates and joint ventures			
Consumer Banking/Wealth Management	0	0	0
Institutional Banking	0	0	5
Treasury Markets	0	3	5
Others	158	150	148
Total	158	153	158
Profit before tax			
Consumer Banking/Wealth Management	1,423	1,993	3,150
Institutional Banking	2,780	4,172	5,022
Treasury Markets	642	421	64
Others	912	357	263
Total	5,757	6,943	8,500

Sources: Compiled from DBS Annual Report 2022 and DBS Annual Report 2023.

Note: Original figures were reported in Singapore Dollars (SGD) and have been converted into U.S. Dollars (US\$) based on the exchange rate: SGD1 = US\$0.74.

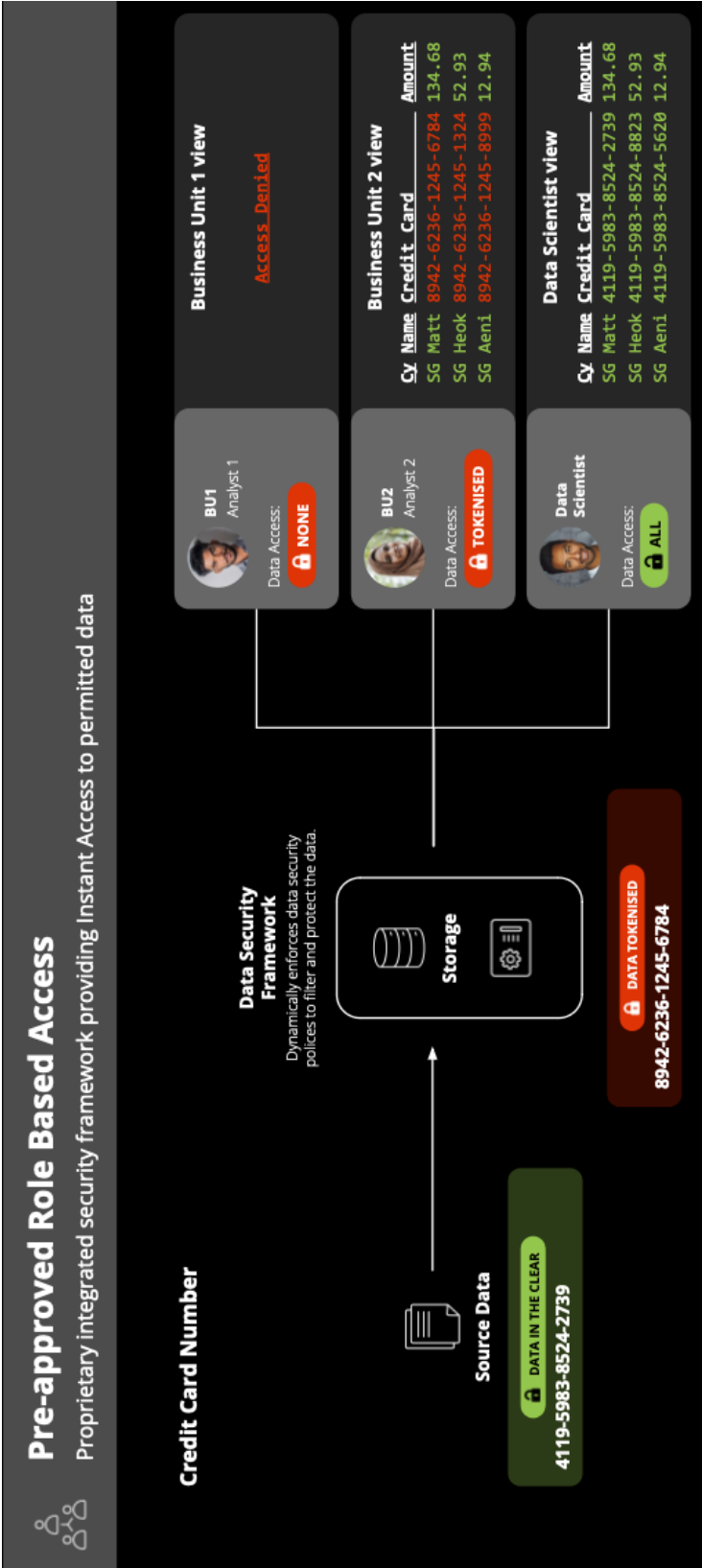
Exhibit 2 Make-up of a Typical Performance Cell

Source: DBS Material.

Exhibit 3 Elements of DBS' MtJ

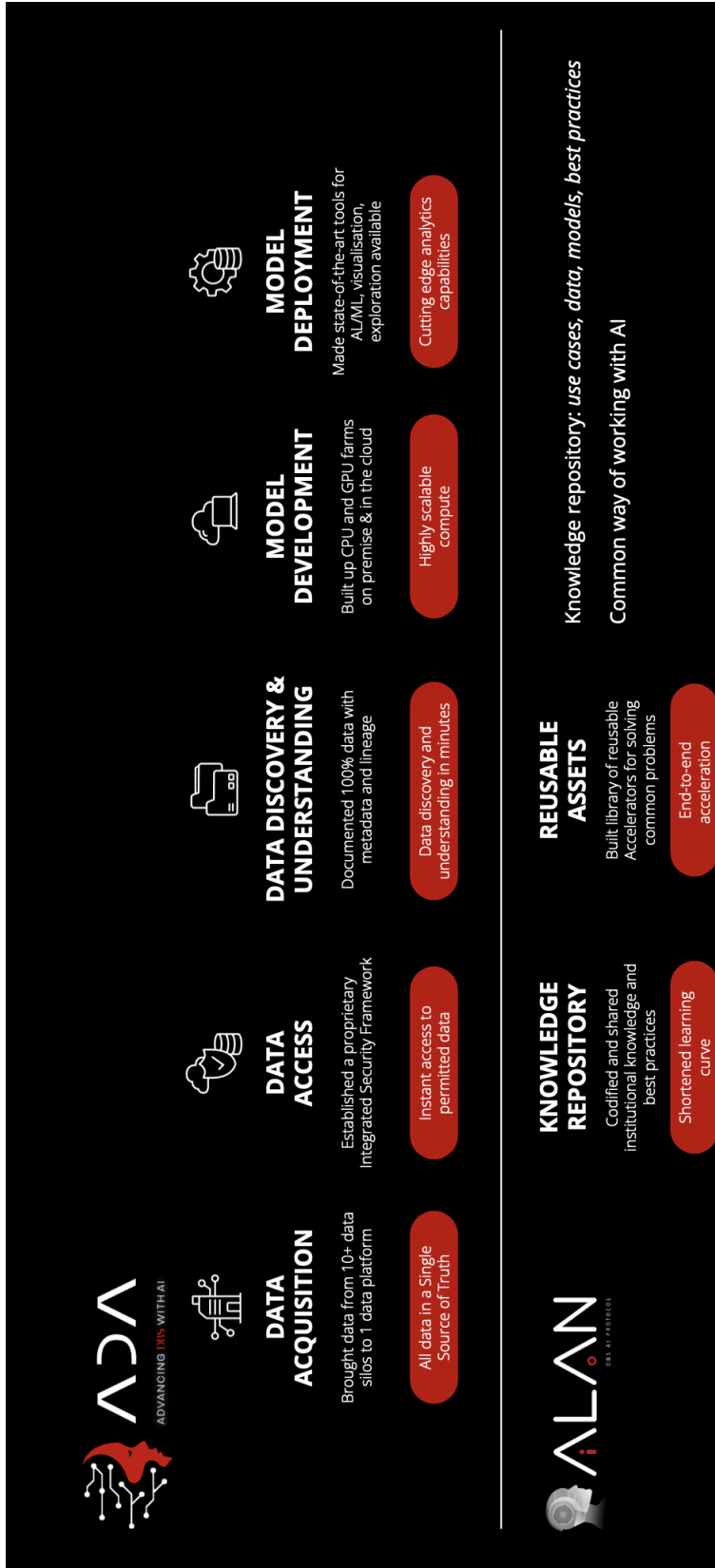
Source: DBS Material.

Exhibit 4 Granting of Data Access by Roles



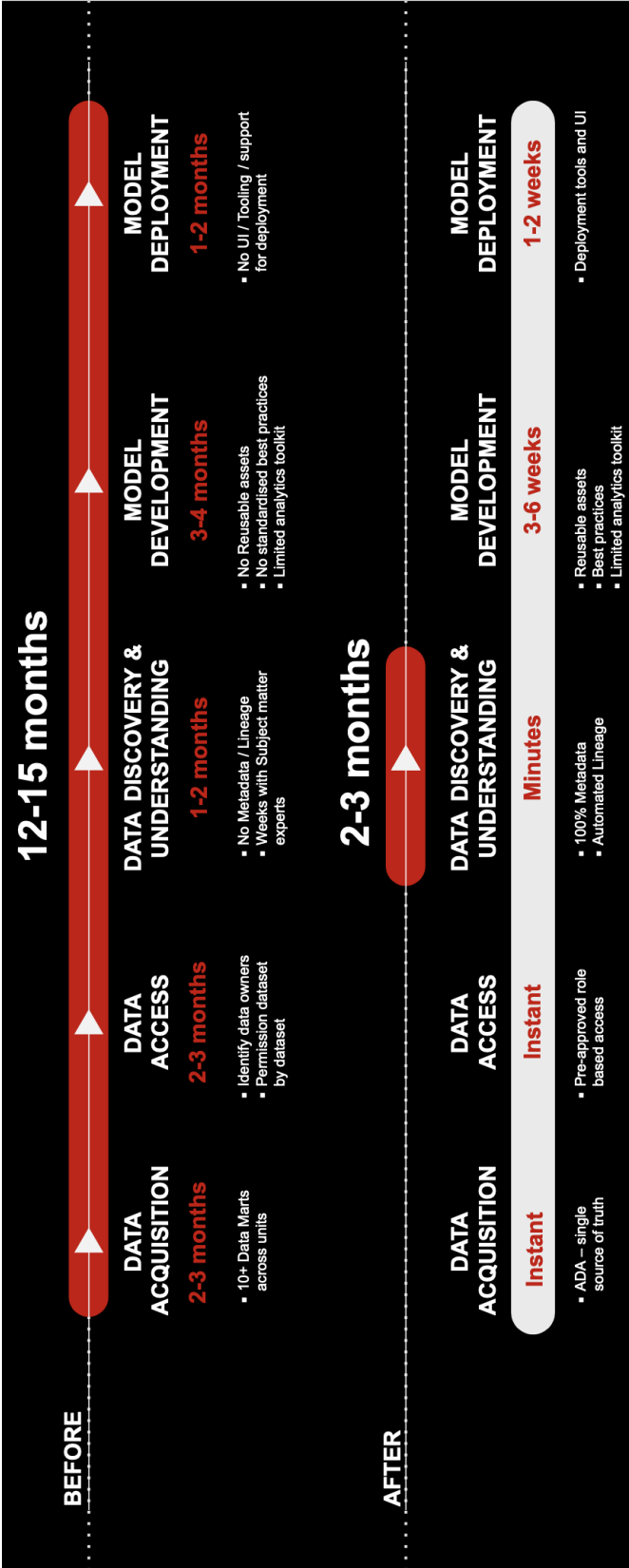
Source: DBS Material.

Exhibit 5 Summary of Key Capabilities of ADA and ALAN



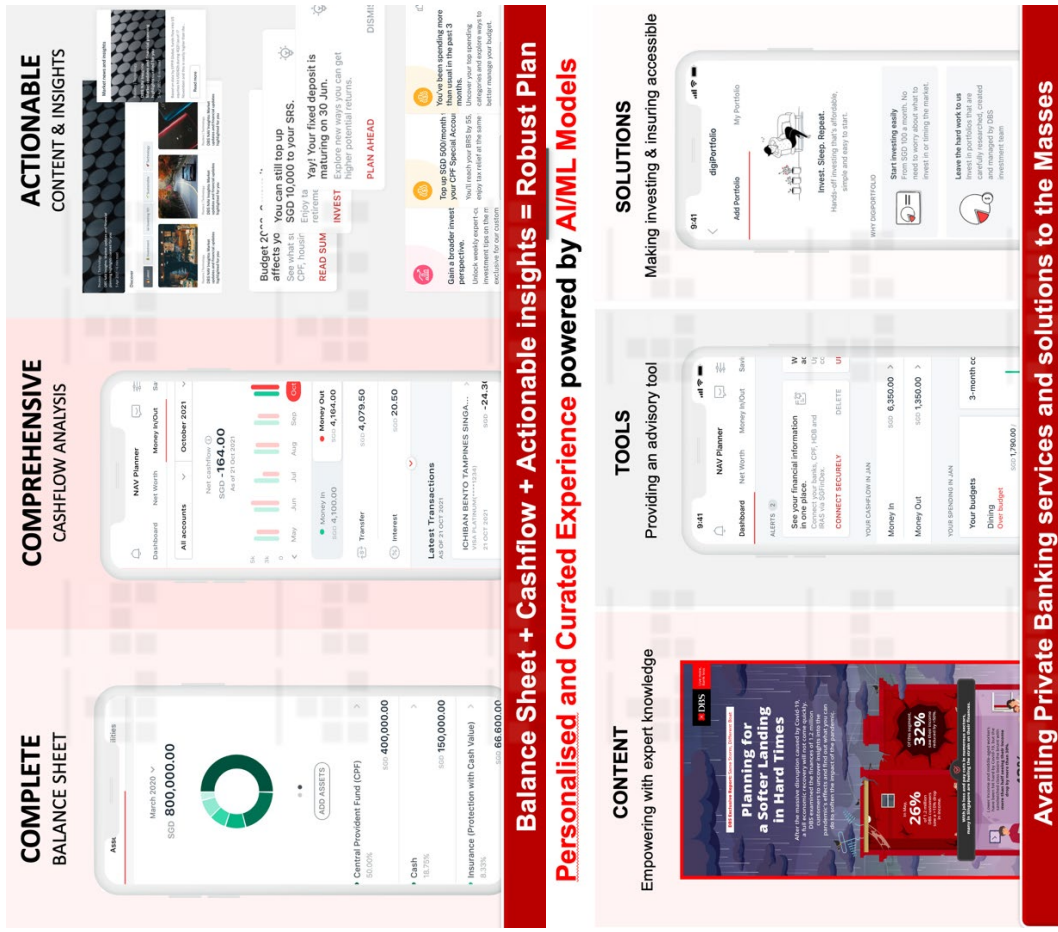
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Exhibit 6 Comparison of Use Case Deployment Timelines



Source: DBS Material.

Exhibit 7 Screenshots of DBS' NAV Planner



Source: DBS Material.

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